

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404004
Kind Code	B1
Date of Patent	September 02, 2025
Inventor(s)	Toš; Patrik et al.

Wing-fuselage joint

Abstract

A wing fuselage joint for an aircraft having joining members which include a first configuration and a second configuration. The joining members of the first configuration include a platform and bracket configured to pivot in a lateral plane and translate in a longitudinal direction. The joining members of the second configuration include a platform and bracket configured to pivot in a longitudinal plane and translate in a lateral direction. The joining members of either configuration join the fuselage structure and the spar of an aircraft together.

Inventors:	Toš; Patrik (Puče, SI), Čretnik; Gregor (Ljubljana, SI)
Applicant:	Pipistrel d.o.o. (Ajdovščina, SI)
Family ID:	94820597
Assignee:	Pipistrel d.o.o. (Ajdovscina, SI)
Appl. No.:	18/591248
Filed:	February 29, 2024

Publication Classification

Int. Cl.: B64C1/26 (20060101); F16B5/02 (20060101)

U.S. Cl.:

CPC B64C1/26 (20130101); F16B5/02 (20130101);

Field of Classification Search

CPC: B64C (1/26); F16B (5/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7887009	12/2010	Keeler, Jr.	244/119	B64C 1/26
9663225	12/2016	Kooiman et al.	N/A	N/A
9981750	12/2017	Williams et al.	N/A	N/A
10040562	12/2017	Kooiman et al.	N/A	N/A
2010/0155532	12/2009	Ariza Martin	244/119	B64C 1/26
2011/0266398	12/2010	Voss	244/45R	B64C 1/26
2022/0017205	12/2021	Vingiani	N/A	B64C 1/26
2022/0048610	12/2021	Diep	N/A	B64C 1/26

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3118104	12/2016	EP	N/A

OTHER PUBLICATIONS

Extended European Search Report issued May 12, 2025 in EP Application No. 25159857.9. cited by applicant

Primary Examiner: Abell; Tye William

Attorney, Agent or Firm: Avek IP, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) None.

BACKGROUND OF THE INVENTION

1. Field

(2) The disclosed embodiments relate generally to the field of aircraft. More specifically, the disclosed embodiments relate to the field of aircraft wing and fuselage connections.

2. Description of the Related Art

(3) It is known to use pin joints to attach a wing to a fuselage on an aircraft. For example, in U.S. Pat. No. 7,887,009 to Keeler et al. the wing is attached to the fuselage using pin joints designed to transfer loads from the wings to the fuselage.

(4) It is also known for a system to mount pylon assemblies in a tiltrotor aircraft. For example, in U.S. Pat. No. 10,040,562 to Kooiman et al. and U.S. Pat. No. 9,981,750 to Williams et al. describe similar systems using spherical bearings to secure pylons to a fuselage or for mounting other major aircraft components to an aircraft.

SUMMARY

(5) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

(6) In some embodiments, the techniques described herein relate to a wing-fuselage joint for an

aircraft including: a first joining member, the first joining member including a first bracket pivotally mounted to a first platform, wherein the first platform is configured to mechanically couple to a fuselage structure and the first bracket is configured to mechanically couple to a wing spar; the first platform being configured to pivot in a first plane and translate in a second direction, wherein the first plane is lateral and aligned with the wing spar and the second direction is perpendicular in relation to the wing spar; a second joining member, the second joining member including a second bracket pivotally mounted to a second platform, wherein the second platform is configured to mechanically couple to the fuselage and the second bracket is configured to mechanically couple to the wing spar; and the second platform being configured to pivot in a second plane and translate in a first direction, wherein the second plane is longitudinal and aligned with the second direction and the first direction is lateral and aligned with the first plane.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

- (1) Illustrative embodiments are described in detail below with reference to the attached drawing figures, which are incorporated by reference herein and wherein:
- (2) FIG. 1 shows a perspective view of the wing-fuselage joint mounted to an aircraft structure;
- (3) FIG. 2 shows a perspective view of a first joining member of the wing-fuselage joint;
- (4) FIG. 3A shows a perspective view of a mounting platform of the first joining member of FIG. 2 mounted to a fuselage structure;
- (5) FIG. 3B shows a cross-sectional view of the first joining member mounted to a spar of a wing;
- (6) FIG. 3C shows an upper perspective view of the mounting platform of the first joining member mounted to the fuselage structure;
- (7) FIG. 3D shows a partially cross-sectional view of the first joining member secured to the spar and fuselage structure;
- (8) FIG. 3E shows a cross sectional view of the first joining member pivotally mounted to a plane bearing;
- (9) FIG. 4 shows a perspective view of a second joining member of the wing-fuselage joint;
- (10) FIG. 5A shows a perspective view of a mounting platform of the second joining member of FIG. 4 mounted to a fuselage structure;
- (11) FIG. 5B shows a cross-sectional view of the second joining member mounted to a spar of a wing;
- (12) FIG. 5C shows an upper perspective view of the mounting platform of the second joining member mounted to the fuselage structure;
- (13) FIG. 5D shows a partially cross-sectional view of the second joining member secured to the spar and fuselage structure;
- (14) FIG. 5E shows a cross sectional view of the second joining member pivotally mounted to a plane bearing;
- (15) FIG. 6 shows a perspective view of an embodiment of the first bracket of FIG. 2 and the second bracket of FIG. 4;
- (16) FIG. 7 shows another perspective view of an embodiment of the first bracket of FIG. 2 and the second bracket of FIG. 4;
- (17) FIG. 8A shows a partially internal view of the first joining member of FIG. 2;
- (18) FIG. 8B shows a partially internal view of the second joining member of FIG. 4, and;
- (19) FIG. 8C shows a perspective view of the plane bearing of FIG. 3E and FIG. 5E.
- (20) The drawing figures do not limit the invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION

(21) The following detailed description references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

(22) In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments but is not necessarily included. Thus, the technology can include a variety of combinations and/or integrations of the embodiments described herein.

(23) Embodiments disclosed herein provide an arrangement for having an improved wing-fuselage joint on an aircraft. Connectors with high strength used for joining a wing and a fuselage together on an aircraft are essential for an aircraft structure. In current arrangements, a wing-fuselage joint may include a bolted joint connection which has high strength, but may have flexibility restrictions, and may require highly complex tooling for manufacture, increasing the cost and time of production. Additionally, specialized analysis is required for current wing-fuselage connections due to the presence of stress concentrations. The connectors must also be able to be rapidly assembled or disassembled to enable transportation of the aircraft. A wing-fuselage joint is needed which may be assembled quickly while also being able to withstand various types of loads such as aerodynamic, ground reaction, vibration, shock, and thermal.

(24) Embodiments disclosed herein, include a wing-fuselage joint for connecting the wing and fuselage structures of an aircraft. The wing-fuselage joint in embodiments comprises spherical bearings and supporting structures having sufficient strength to withstand numerous different types of loading. The wing-fuselage joint transfers the shear forces experienced by an aircraft's wings in numerous directions to the fuselage of an aircraft while substantially preventing the bending moment from being transferred to the fuselage structure. The reduction or elimination of the bending moment to the fuselage allows for the fuselage structure size, complexity, cost, and manufacture time, to be reduced.

(25) FIG. 1 shows a wing-fuselage joint **100** mounted onto an aircraft structure in an embodiment. The wing-fuselage joint **100** is configured for mounting a fuselage structure **102** with a wing structure or spar **104**. The fuselage structure is for example a girder or frame station. The fuselage structure **102** and the spar **104** provide structural support for the fuselage and the wings of an aircraft respectively. The wing-fuselage joint **100** is mounted to members of both the fuselage structure **102** and the spar **104** and mechanically couples the fuselage structure **102** and spar **104** together. The wing-fuselage joint **100** includes four joining members **106** (three may be seen in FIG. 1). The joining members **106** include components to attach and secure to the fuselage structure **102** and spar **104** in four separate locations in embodiments. The joining members may be placed on a forward spar, aft spar, a forward bulkhead, or an aft bulkhead. The joining members **106** include two different configurations with two joining members **106** having a first configuration and another two joining members having a second configuration.

(26) In embodiments and with reference to FIG. 1, the X direction is defined as a longitudinal direction aligned forward and aft along the length of the fuselage, the Y direction is a lateral direction that spans left and right across the aircraft in the main direction of the wings, and the Z direction is a vertical direction aligned up and down. The first joining members **108** having the first

configuration are configured to transfer forces experienced by the wings or spar **104** to the fuselage structure **102** in the Y and Z directions, and the second joining members **110** having the second configuration are configured to transfer forces experienced by the wings or spar **104** to the fuselage structure **102** in the X and Z directions.

(27) FIG. 2 shows a close-up perspective view of one of the first joining members **108**. Each first joining member **108** includes a mounting platform **208**. The mounting platform **208** is substantially square shaped in embodiments and includes a center aperture **213** removed from a middle area of the mounting platform **208**. At each of the four corners of the mounting platform **208**, a bolt and nut arrangement **214** is inserted into a fastener hole in the mounting platform **208**. The bolt and nut arrangement **214** is configured such that the bolt head is positioned on the underside surface of the mounting platform **208** such that the shaft of the bolt extends upwards and through the fastener hole in the mounting platform **208**. As each bolt shaft extends upwards one end of a fastener plate **216** is slid onto the bolt shaft with a nut being threadingly secured onto the upper end of the bolt shaft. The mounting platform **208** includes two fastener plates **216** with one fastener plate **216** being inserted on one end of the forward-right bolt and nut arrangement **214** and on the other end being inserted onto the back-right bolt and nut arrangement **214**. Another fastener plate **216** is inserted onto the forward-left bolt and nut arrangement **214** and on the other end is inserted onto the back-left bolt and nut arrangement **214**. The fastener plates **216** span from the forward to the back edges of the mounting platform **208** and are positioned along the left and right sides of the mounting platform **208**.

(28) FIG. 3A shows a bottom perspective view of the mounting platform **208** mounted to the fuselage structure **102**. The mounting platform **208** includes two downwardly extending flanges **210** which each include a fastener hole **211** diametrically sized to allow a shear pin **207** (FIG. 3D) to insert and couple the mounting platform **208** to a plane bearing **112**. The flanges **210** are spaced opposite each other and mounted on the underside of the mounting platform **208** such that a first bracket **202** (FIG. 3D) is able to pivot on the plane bearing **112** in between the flanges **210**.

(29) With reference to FIG. 2 and FIG. 3D, the first bracket **202** has an A shape but could have an L shape, T shape, triangle shape, or other shape without departing from the scope hereof. The first bracket **202** includes a grease nipple **105** so that grease may be applied to grease within the first bracket **202**. With reference to FIGS. 6 and 7, the first bracket **202** may be slid onto lamination pins **204** laminated into holes in the spar **104** and positioned such that an end of a pin bolt **206** may be inserted and screwed into the wing lamination pin **204**. The wing lamination pins **204** include a lamination nut **203** and a pin spacer **205** configured with the pin bolt **206**. The wing lamination pins **204** are laminated into the spar **104** structure which may eliminate the need for elements such as rods to keep the wing lamination pins **204** in place. The mounting platform **208** is pivotally attached at the center of the first bracket **202** using a plane bearing **112** (FIGS. 8A and 8B) fit within the first bracket **202**. The fastener holes **211** (one is shown in FIG. 3D) and the wing lamination pins **204** are both aligned with the X-Z plane (the X-Z plane being longitudinal and aligned with the X and Z direction). A shear pin **207** is configured to insert through the aligned fastener holes **211** (FIG. 3E).

(30) FIG. 3B shows a cross sectional view of the first joining member **108** mounted onto the spar **104**. The first joining member **108** is secured to the spar **104** by being slid over a wing lamination pin **204** prior to the insertion of the pin bolts **206**. The pin bolts **206** are configured to screw directly into the lamination pins **204**. In embodiments, a single wing lamination pin **204** is shown adjacent to the pin bolt **206** inserting through the spar **104**.

(31) FIG. 3C shows a top perspective view of the first joining member **108** being secured to the fuselage structure **102** using the bolt and nut arrangement **214** and the fastener plate **216**. Each of the four bolt of the bolt and nut arrangements **214** insert through formed holes in the fuselage structure **102** such that the fastener plate **216** on either end of the mounting platform **208** are each positioned on the surface of the fuselage structure **102**.

(32) FIG. 3D shows the nut threadingly secured on the end of the bolt above the fastener plate **216**. The fuselage structure **102** is sandwiched in between the mounting platform **208** and the fastener plate **216** which substantially secures the first joining member **108** to the fuselage structure **102**. The mounting platform **208** is rigidly attached to fuselage structure **102** and the plane bearing **112** couples the first bracket **202** to the mounting platform **208** allowing the first bracket **202** to pivot around plane bearing **112** along the Y-Z plane while having the ability to translate in the X direction. This substantially allows for the first joining member **108** to transfer forces experienced by the spar **104** to the fuselage structure **102**. For instance, when the spar **104** experiences a force or forces in the Y or Z direction the first joining member **108** allows for the force or forces to be substantially transferred to the fuselage structure **102**. A force experienced by the spar **104** may be from an aerodynamic, ground reaction, vibration, or shock load applied to the aircraft on the ground or in flight.

(33) FIG. 3E shows a cross sectional view of the first joining member **108**. The flanges **210** extend downwards and are longitudinally spaced apart such that the plane bearing **112** is able to be positioned in between the flanges **210** pivotally coupling the first bracket **202** to the mounting platform **208**. The first joining member **108** is configured such that the plane bearing **112** allows the mounting platform **208** to pivot freely and translate in the X direction relative to the first bracket **202**. The shear pin **207** is slid through an inner rim **402** of the plane bearing **112** (FIG. 8C) and allows for the first joining member **108** to translate in the X direction due to the clearance between the plane bearing **112** within the first bracket **202** positioned in between the flanges **210**. In embodiments, bearings **109** may be press fit into the first bracket **202** such that the shear pins **207** slide through. The bearings **109** may possibly provide rotatability for the shear pins **207** inserted through the first bracket **202**.

(34) FIG. 4 shows a close-up perspective view of one of the second joining member **110**. Each second joining member **110** includes a mounting platform **308**. The mounting platform **308** is substantially square shaped in embodiments. At each of the four corners of the mounting platform **308**, a bolt and nut arrangement **214** is inserted into a fastener hole in the mounting platform **308**. The bolt and nut arrangement **214** is configured such that the bolt head is positioned on the underside surface of the mounting platform **308** such that the shaft of the bolt extends upwards and through the fastener hole in the mounting platform **308**. As each bolt shaft extends upwards one end of a fastener plate **316** is slid onto the bolt shaft with a nut being threadingly secured onto the upper end of the bolt shaft. The mounting platform **308** includes two fastener plates **316** which are inserted onto the bolt and nut arrangement **214** similarly to the fastener plate **216** configured with the mounting platform **208**. One fastener plate **316** is inserted on one end of the forward-right bolt and nut arrangement **214** and on the other end is secured onto the forward-left bolt and nut arrangement **214**. Another fastener plate **316** is inserted onto the back-right bolt and nut arrangement **214** and on the other end is inserted onto the back-left bolt and nut arrangement **214**. With reference to FIG. 4, the fastener plates **316** span from the right to the left edges of the mounting platform **308** being positioned along the forward and back sides of the mounting platform **308**.

(35) FIG. 5A shows a bottom perspective view of the mounting platform **308** mounted to the fuselage structure **102**. The mounting platform **308** includes two downwardly extending flanges **310** which each include a fastener hole **211** diametrically sized to allow a shear pin **207** to insert and couple the mounting platform **208** to a plane bearing **112** (FIG. 3D). The flanges **310** are spaced opposite each other and mounted on the underside of the mounting platform **208** such that a second bracket **302** (FIG. 3D) is able to pivot on the plane bearing **112** in between the flanges **310**.

(36) With reference to FIG. 4 and FIG. 5D, the second bracket **302** which, in embodiments, includes three extending arms. Two extending arms of the second bracket **302** extend downwards with reference to FIG. 4, with each arm end being slide through a wing lamination pin **304** (also shown in FIG. 5B) securing an end of a pin bolt **206** to each downwardly extending arm. The wing

lamination pins **304** include a lamination nut **203** and a pin spacer **205** configured with the pin bolt **206**. The wing lamination pins **304** are laminated into the spar **104** structure which may eliminate the need for elements such as rods to keep the wing lamination pins **304** in place. The mounting platform **308** is pivotally attached at the center of the second bracket **302** using a plane bearing **112** (FIGS. **8A** and **8B**) press fit within the second bracket **302**. The fastener holes **311** (one is shown in FIG. **3D**) are aligned in the Y-Z plane perpendicular to the wing lamination pins **304** aligned in the X-Z plane (the Y-Z plane being lateral and aligned with the Y and Z direction). A shear pin **307** is configured to insert through the aligned fastener holes **311**.

(37) FIG. **5B** shows a cross sectional view of the second joining member **110** mounted onto the spar **104**. The second joining member **110** is secured to the spar **104** using pin bolts **206** which screw directly into wing lamination pins **304** laminated into the spar **104**. In embodiments, two wing lamination pins **304** are shown adjacent to the pin bolts **206**.

(38) FIG. **5C** shows a top perspective view of the second joining member **110** being secured to the fuselage structure **102** using the bolt and nut arrangement **214** and the fastener plate **316**. Each of the four bolt of the bolt and nut arrangements **214** insert through formed holes in the fuselage structure **102** such that the fastener plate **216** on either end of the mounting platform **208** are each positioned on the surface of the fuselage structure **102**.

(39) FIG. **5D** shows the nut threadingly secured on the end of the bolt above the fastener plate **316**. The fuselage structure **102** is sandwiched in between the mounting platform **308** and the fastener plate **316** which substantially secures the second joining member **110** to the fuselage structure **102**. The mounting platform **308** is rigidly attached to fuselage structure **102** and the plane bearing **112** couples the second bracket **302** to the mounting platform **308** allowing the second bracket **302** to pivot around plane bearing **112** along the X-Z plane while having the ability to translate in the Y direction. This substantially allows for the first joining member **108** to transfer forces experienced by the spar **104** to the fuselage structure **102**. For instance, when the spar **104** experiences a force or forces in the X or Z direction the second joining member **110** allows for the force or forces to be substantially transferred to the fuselage structure **102**. A force experienced by the spar **104** may be from an aerodynamic, ground reaction, vibration, or shock load applied to the aircraft on the ground or in flight.

(40) FIG. **5E** shows a cross sectional view of the second joining member **110**. The flanges **310** extend downwards and are longitudinally spaced apart such that the plane bearing **112** is able to be positioned in between the flanges **310** pivotally coupling the second bracket **302** to the mounting platform **308**. The second joining member **110** is configured such that the plane bearing **112** allows the mounting platform **308** to pivot freely and translate in the Y direction relative to the second bracket **302**. The shear pin **307** is slid through an inner rim **402** of the plane bearing **112** (FIG. **8C**) and allows for the second joining member **110** to translate in the Y direction due to the clearance between the plane bearing **112** fit within the second bracket **302** positioned in between the flanges **310**. In embodiments, bearings **109** may be press fit into the second bracket **202** such that the shear pins **307** slide through. The bearings **109** may possibly provide rotatability for the shear pins **307** inserted through the second bracket **302**.

(41) FIG. **6** shows a perspective view of an embodiment with the second bracket **302** forward of the first bracket **202** each including the wing lamination pins **204**, pin bolts **206** and the lamination nuts **203**. When coupled to the wing structure, the pin bolts **206** are screwed into the wing lamination pins **204** laminated into holes in the spar **104** and mount the first bracket **202** and second bracket **302** to the spar **104** structure. The first bracket **202** allows the mounting platform **208** to freely rotate and translate in the X direction relative to the fuselage structure **102**. This allows for forces experienced by the spar **104** in the Y or Z directions to be transferred to the fuselage structure **102**. The second bracket **302** allows the mounting platform **308** to freely rotate and translate in the Y direction relative to the fuselage structure **102**. This allows for forces experienced by the spar **104** in the X or Z directions to be transferred to the fuselage structure **102**.

The brackets **202** and **302** when attached to the mounting platforms **208** and **308** as shown in FIG. **1** are configured such that the first joining member **108** and the second joining member **110** are positioned on the same side (i.e. left or right) of the fuselage structure **102** which enables forces experienced by the spar **104** in the X and Y directions to be substantially reduced due to pivoting or translation accomplishable by both the first joining member **108** and the second joining member **110**. Since both the first joining member **108** and the second joining member **110** allow for free rotation about the plane bearing **112** in either the Y-Z plane or the X-Z plane, a moment load applied in the Z direction to the spar **104** would be minimally transferred or eliminated from travelling to the fuselage structure **102**. In embodiments, when fully assembled as shown in FIG. **1**, the diagonal configuration of the first joining member **108** and the second joining member **110** allow for local translations in either the X direction or the Y direction (i.e. first joining member **108** allows X direction translation and second joining member **110** allows Y direction translation) substantially enabling the wing to bend and deform with minimum restraint and the reaction forces on the wing spars **104** and on the fuselage structure **102** to be reduced. This may include substantially eliminating the bending moment applied from the lifting force to the wings from being transferred to the fuselage of an aircraft. The first joining member **108** and the second joining member **110** each are configured to have free rotation in the Y-Z plane or X-Z plane which allows wing displacement caused from upwards bending (i.e. a force applied in the Z direction) to occur.

(42) FIG. **7** shows a perspective view of an embodiment with the first bracket **202** forward of the second bracket **302**. The second bracket **302** when attached to the mounting platform **308** (shown in FIG. **4**) allows the mounting platform **308** to freely pivot in the X-Z plane and translate in the Y direction relative to the fuselage structure **102**. This allows for forces experienced by the spar **104** in the X or Z directions to be transferred to the fuselage structure **102**. The first bracket **202** when attached to the mounting platform **208** (shown in FIG. **1**) allows the mounting platform **208** to freely pivot in the Y-Z plane and translate in the X direction relative to the fuselage structure **102**. This allows for forces experienced by the spar **104** in the Y or Z directions to be transferred to the fuselage structure **102**. Both the first bracket **202** and second bracket **302** when attached to the mounting platform **208** and mounting platform **308** allow the first joining member **108** and the second joining member **110** to be positioned on the same side (i.e. left or right side) of the fuselage structure **102** which enables forces experienced by the spar **104** in the X and Y directions to be substantially reduced due to pivoting accomplishable by both the first joining member **108** and the second joining member **110**. Since both the first joining member **108** and the second joining member **110** are able to freely pivot in either the Y-Z plane or the X-Z plane about the plane bearing **112** the wings are able to bend and deform with minimal restraint while reaction forces on the fuselage structure **102** and spar **104** are significantly reduced.

(43) FIGS. **8A** and **8B** provide an internal view of the first joining member **108** and the second joining member **110** respectively. The first joining member **108** and the second joining member **110** each include a plane bearing **112**. FIG. **8C** shows the plane bearing **112** which includes an inner rim **402** and an outer rim **404** with each rim having a common center. The outer rim **404** is configured to rotate independently of the inner rim **402**. With respect to FIG. **8A**, the rotation of the plane bearing **112** allows for the mounting platform **208** to pivot in the Y-Z plane relative to the first bracket **202**. With respect to FIG. **8B**, the rotation of the plane bearing **112** allows for the mounting platform **308** to pivot in the X-Z plane relative to the second bracket **302**. In embodiments, the plane bearing **112** may be an SKF™ GE_30_ESX_2LS type bearing which may be press fit into either the first bracket **202** or second bracket **302**. The inner rim **402** of the plane bearing **112** has a diameter which allows the shear pin **207** or the shear pin **307** to insert and slide into the plane bearing **112** so the first joining member **108** is able to translate in the X direction and the second joining member **110** is able to translate in the Y direction. For instance, when installed onto the fuselage structure **102** and spar **104** the first joining member **108** is configured to allow the wing to pivot in the Y-Z plane and not translate in the Y and Z direction while the second joining member

110 is configured to allow the wing to pivot in the X-Z plane and not translate in the X and Z direction. This allows the wing-fuselage joint **100** to be flexible while providing strength simultaneously.

(44) The joining members **106** may be arranged in strategic ways and locations about the aircraft fuselage **102** and the wing spar **104** to create an optimal wing-fuselage joint **100**. The joining members **106** may be arranged based upon the specific aircraft wing-fuselage configuration, load paths, installation, production tolerance, and possible disassembly requirements. For instance, with respect to FIG. 1, the first joining members **108** and the second joining members **110** may be switch frontward and rearward and leftward and rightward based upon the load path of the aircraft or another characteristic apparent to one skilled in the art. The wing-fuselage joint **100** may be implemented on numerous different types of aircraft including commercial and military airplanes, helicopters, vertical take off and landing aircraft, and unmanned aerial vehicles.

(45) The wing-fuselage joint **100** is able to substantially reduce various types of loading which may be experienced by an aircraft such as aerodynamic loads, ground reaction loads, wing mounted propeller lifting loads, vibration loads, shock loads, and thermal loads which may be reduced by free planar rotation and X or Y translation of the mounting platform **208** and mounting platform **308** mounted to the spar **104** of the aircraft. Additionally, the wing-fuselage joint **100** including the plane bearing **112** and securing arrangement connections allow for a quicker assembly and disassembly process than current methods. The wing-fuselage joint **100** allows for greater production tolerance errors while reducing the production tool cost and complexity due to its translatability and rotatability. The configuration of the wing-fuselage joint **100** allows for minor movements and deformations of the aircraft to have little impact on the overall performance of the aircraft. The wing-fuselage joint **100**, in embodiments, includes four joining members **106** of the first configuration or second configuration which each secure the fuselage structure **102** and the spar **104** structures together. In this configuration, the wing-fuselage joint **100** is able to claim fail-safe design characteristics due to each of the joining members **106** being able to possess significant load carrying capacity in case instances where one or more joining members **106** fail.

(46) Features described above as well as those claimed below may be combined in various ways without departing from the scope hereof. The following clauses illustrate some possible, non-limiting combinations:

(47) Clause 1. A wing-fuselage joint for an aircraft comprising: a first joining member, the first joining member including a first bracket pivotally mounted to a first platform, wherein the first platform is configured to mechanically couple to a fuselage structure and the first bracket is configured to mechanically couple to a wing spar; the first platform being configured to pivot in a first plane and translate in a second direction, wherein the first plane is lateral and aligned with the wing spar and the second direction is perpendicular in relation to the wing spar; a second joining member, the second joining member including a second bracket pivotally mounted to a second platform, wherein the second platform is configured to mechanically couple to the fuselage and the second bracket is configured to mechanically couple to the wing spar; and the second platform being configured to pivot in a second plane and translate in a first direction, wherein the second plane is longitudinal and aligned with the second direction and the first direction is lateral and aligned with the first plane.

(48) Clause 2. The wing fuselage joint of clause 1 wherein a forward first joining member mechanically couples a forward spar and a forward bulkhead and an aft first joining member mechanically couples an aft spar and an aft bulkhead and a forward second joining member mechanically couples a forward spar and forward bulkhead and an aft second joining member mechanically couples an aft spar and an aft bulkhead.

(49) Clause 3. The wing fuselage joint of clauses 1-2 wherein the forward first joining member and the aft first joining member are diagonally opposed to one another, the forward second joining member and the aft second joining member are diagonally opposed to one another.

(50) Clause 4. The wing fuselage joint of clauses 1-3 wherein the first bracket and the second bracket each include a first extending arm and a second extending arm, wherein the first and second extending arms each are configured to receive a pin bolt.

(51) Clause 5. The wing fuselage joint of clauses 1-4 wherein a pin bolt is secured into the first and second extending arms of the first bracket, and another pin bolt is secured into the first and second extending arms of the second bracket.

(52) Clause 6. The wing fuselage joint of clauses 1-5 wherein wing lamination pins are laminated into the aft spar and the wing lamination pins are laminated into the forward spar.

(53) Clause 7. The wing fuselage joint of clauses 1-6 wherein a lamination nut and a pin spacer are inserted onto each pin bolt.

(54) Clause 8. The wing fuselage joint of clauses 1-7 wherein the first joining member includes a first plane bearing and the second joining member includes a second plane bearing.

(55) Clause 9. The wing fuselage joint of clauses 1-8 wherein the first plane bearing is fit into the first bracket and mechanically couples the first bracket to the first platform and allows the first platform to pivot in the first plane.

(56) Clause 10. The wing fuselage joint of clauses 1-9 wherein the second plane bearing is fit into the second bracket and mechanically couples the second bracket to the second platform and allows the second platform to pivot in the second plane.

(57) Clause 11. The wing fuselage joint of clauses 1-10 wherein the first platform is secured to the fuselage structure by a first fastener plate and a second fastener plate aligned in the first plane wherein the fuselage structure is sandwiched in between a surface of the first platform and the first and second fastener plates.

(58) Clause 12. The wing fuselage joint of clauses 1-11 wherein the second platform is secured to the fuselage structure by a first fastener plate and a second fastener plate aligned in the second plane wherein the fuselage structure is sandwiched in between a surface of the second platform and the first and second fastener plates.

(59) Clause 13. The wing fuselage joint of clauses 1-12 wherein the first and second fastener plates are secured using a nut and bolt arrangement.

(60) Clause 14. The wing fuselage joint of clauses 1-13 wherein the first platform includes a first pair of oppositely opposed flanges extending away from the first platform and the second platform includes a second pair of oppositely opposed flanges extending away from the second platform.

(61) Clause 15. The wing fuselage joint of clauses 1-14 wherein the first pair of oppositely opposed flanges are opposingly spaced such that the first plane bearing fitted within the first bracket may translate the first bracket in the second direction, and the second pair of oppositely opposed flanges are opposingly spaced such that the second plane bearing fitted within the second bracket may translate the second bracket in the first direction.

(62) Many different arrangements of the various components depicted, as well as components not shown, are possible without departing from the spirit and scope of what is claimed herein. Embodiments have been described with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from what is disclosed. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from what is claimed.

(63) It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims. Not all steps listed in the various figures need be carried out in the specific order described.

Claims

1. A wing-fuselage joint for an aircraft comprising: a first joining member, the first joining member including a first bracket pivotally mounted to a first platform, wherein the first platform is configured to mechanically couple to a fuselage structure and the first bracket is configured to mechanically couple to a wing spar; the first platform being configured to pivot in a first plane and translate in a second direction, wherein the first plane is lateral and aligned with the wing spar and the second direction is perpendicular in relation to the wing spar; a second joining member, the second joining member including a second bracket pivotally mounted to a second platform, wherein the second platform is configured to mechanically couple to the fuselage and the second bracket is configured to mechanically couple to the wing spar; and the second platform being configured to pivot in a second plane and translate in a first direction, wherein the second plane is longitudinal and aligned with the second direction and the first direction is lateral and aligned with the first plane.
2. The wing fuselage joint of claim 1 wherein the first platform is secured to the fuselage structure by a first fastener plate and a second fastener plate aligned in the first plane wherein the fuselage structure is sandwiched in between a surface of the first platform and the first and second fastener plates.
3. The wing fuselage joint of claim 1 wherein a forward first joining member mechanically couples a forward spar and a forward bulkhead and an aft first joining member mechanically couples an aft spar and an aft bulkhead and a forward second joining member mechanically couples a forward spar and forward bulkhead and an aft second joining member mechanically couples an aft spar and an aft bulkhead.
4. The wing fuselage joint of claim 3 wherein the forward first joining member and the aft first joining member are diagonally opposed to one another, the forward second joining member and the aft second joining member are diagonally opposed to one another.
5. The wing fuselage joint of claim 1 wherein the second platform is secured to the fuselage structure by a first fastener plate and a second fastener plate aligned in the second plane wherein the fuselage structure is sandwiched in between a surface of the second platform and the first and second fastener plates.
6. The wing fuselage joint of claim 5 wherein the first and second fastener plates are secured using a nut and bolt arrangement.
7. The wing fuselage joint of claim 1 wherein the first platform includes a first pair of oppositely opposed flanges extending away from the first platform and the second platform includes a second pair of oppositely opposed flanges extending away from the second platform.
8. The wing fuselage joint of claim 7 wherein the first pair of oppositely opposed flanges are opposingly spaced such that the first plane bearing fitted within the first bracket may translate the first bracket in the second direction, and the second pair of oppositely opposed flanges are opposingly spaced such that the second plane bearing fitted within the second bracket may translate the second bracket in the first direction.
9. The wing fuselage joint of claim 1 wherein the first joining member includes a first plane bearing and the second joining member includes a second plane bearing.
10. The wing fuselage joint of claim 9 wherein the first plane bearing is fit into the first bracket and mechanically couples the first bracket to the first platform and allows the first platform to pivot in the first plane.
11. The wing fuselage joint of claim 9 wherein the second plane bearing is fit into the second bracket and mechanically couples the second bracket to the second platform and allows the second platform to pivot in the second plane.
12. The wing fuselage joint of claim 1 wherein the first bracket and the second bracket each include a first extending arm and a second extending arm, wherein the first and second extending arms each are configured to receive a pin bolt.

13. The wing fuselage joint of claim 12 wherein a pin bolt is secured into the first and second extending arms of the first bracket, and another pin bolt is secured into the first and second extending arms of the second bracket.
 14. The wing fuselage joint of claim 13 wherein wing lamination pins are laminated into the aft spar and the wing lamination pins are laminated into the forward spar.
 15. The wing fuselage joint of claim 13 wherein a lamination nut and a pin spacer are inserted onto each pin bolt.
-